



12700 - The first NIRSpec observations of a relativistic tidal disruption event candidate

Cycle: 4, Proposal Category: DD

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		NIRSpec Fixed Slit Spectroscopy	(1) EP260302a

ABSTRACT

We request a NIRSpec fixed slit observation of EP260302a, a newly discovered multiwavelength transient with properties that are consistent with a relativistic tidal disruption event (rTDE). These systems, produced by the disruption of a star by a massive black hole, drive exceptionally powerful relativistic jets and are vanishingly rare. EP260302a is only the fifth candidate and offers the first opportunity for JWST to observe such a source.

However, the collapse of massive stars can also produce a transient with similar properties to rTDEs. Our proposed observation will, for the first time, provide definitive evidence of the progenitor system of an rTDE candidate through spectroscopic identification of features linking it to either a TDE or collapsar origin. This will have profound implications for the sample as a whole and mark a significant step forward in our understanding of these rare and extreme transient events.

OBSERVING DESCRIPTION

We request a NIRSpec fixed slit observation of EP260302a, a newly discovered multiwavelength transient with properties that are consistent with a relativistic tidal disruption event (rTDE). These systems, produced by the disruption of a star by a massive black hole, drive exceptionally powerful relativistic jets and are vanishingly rare. EP260302a is only the fifth candidate and offers the first opportunity for JWST to observe such a source. However, the collapse of massive stars can also produce a transient with similar properties to rTDEs. Our proposed observation will, for the first time, provide definitive evidence of the progenitor system of an rTDE candidate through spectroscopic identification of features linking it to either a TDE or collapsar origin. This will have profound implications for the sample as a whole and mark a significant step forward in our understanding of these rare and extreme transient events.

Proposal 12700 - Targets - The first NIRSpect observations of a relativistic tidal disruption event candidate

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	EP260302a	RA: 08 51 53.6915 (132.9737146d) Dec: -65 59 0.97 (-65.98360d) Equinox: J2000	Epoch of Position: 2016.0	
	Comments: Coordinates derived using optical imaging calibrated to Gaia DR3. Category=Unidentified Description=[Infrared sources, Radio sources, Variable radiation sources, Visible sources, X-ray sources] Extended=NO				
Fixed Targets	(2)	2MASS08515168-6559092	RA: 08 51 51.6457 (132.9651904d) Dec: -65 59 8.93 (-65.98581d) Equinox: J2000	Parallax: 0.0007973" Epoch of Position: 2016.0	
	Comments: This object was generated by the targetselector and retrieved from the 2MASS database. The coordinates have been adjusted based on our optical imaging using the same method and frames as the target (EP260302a) to minimise errors in the offset. While the target does have some proper motion, it should be negligible over the time frame between the optical observations and JWST observing.				
	Category=Star Description=[K dwarfs, K stars]				

Proposal 12700 - Observation 1 - The first NIRSpect observations of a relativistic tidal disruption event candidate

Observation	Proposal 12700, Observation 1										Tue May 26 20:00:48 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRSpect Fixed Slit Spectroscopy											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	EP260302a	RA: 08 51 53.6915 (132.9737146d)			Epoch of Position: 2016.0						
			Dec: -65 59 0.97 (-65.98360d)									
			Equinox: J2000									
		Comments: Coordinates derived using optical imaging calibrated to Gaia DR3. Category=Unidentified Description=[Infrared sources, Radio sources, Variable radiation sources, Visible sources, X-ray sources] Extended=NO										
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	2 2MASS08515168 -6559092	WATA	SUB32	F140X	NRSRAPID	3	1	1	0.08	303094	
Template	HFF Readout Mode			Slit			Subarray					
	false			S400A1			FULL					
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	3						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	S400A1	NRSIRS2RAPID	51	1	1	NONE	3	3	2275.867	303094

Proposal 12700 - Observation 1 - The first NIRSpec observations of a relativistic tidal disruption event candidate

Special Requirements	Before Date 15-JUN-2026:00:00:00
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